

Tobechukwu David Uka

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PROFESSIONAL SUMMARY

Senior Backend Engineer (6+ years) focused on real-time systems and payments. I design low-latency, event-driven backends, build fault-tolerant APIs and job pipelines, and keep systems observable and incident-ready. Strong in rate limiting, idempotency, and secure payment flows; comfortable owning architecture and collaborating with leadership.

TECHNOLOGIES

Back-End & Cloud	Node.js/TypeScript, Express/NestJS, Python, C#, .NET, Java, Microservices, GraphQL, Spring Boot, Docker, Kafka, RabbitMQ, Azure, AWS, WebSockets, SQS/SNS, Kafka/RabbitMQ, API Gateway, Lambda, MongoDB, PostgreSQL, EC2, CloudWatch, CloudFront
AI & Machine Learning	LLMs; SFT; RL; prompt design; AI agents (tool/API calling); Retrieval-Augmented Generation (RAG); Embeddings and Vector Search (pgvector, Pinecone); Chunking and Indexing; Reranking; Function/Tool Calling; Benchmark Design; Schema-validated outputs (Zod/JSON Schema)
Front-End	ReactJS, NextJS, Angular, JavaScript, TypeScript, HTML5, CSS3
DevOps & Agile	CI/CD pipelines, Git, Code Reviews, Scrum, Automated Testing
Leadership	Team Building, Mentorship, Architectural Design, Stakeholder Engagement

EXPERIENCE

Senior Software Engineer (POD LEAD) | Turing | California, US | Nov 2024 – Present

- Owned backend services in Node.js/TypeScript for AI/ops tooling; designed pub/sub event streams and job workers for high-throughput tasks.
- Implemented connection management and backpressure strategies; optimized hot paths to improve p95/p99 latency and reduce tail timeouts.
- Productionized CI/CD with staged rollouts and health probes; added structured logs, traces, and SLO dashboards.
- Partnered with leadership on technical direction; wrote ADRs and set engineering standards (reviews, testing, runbooks).
- Conduct model red-teaming and add guardrails (safety filters and output validation) to reduce harmful or off-policy responses.
- Partner with product and platform teams to scope use cases, manage cost/performance, and ship reliable model updates on a steady cadence.
- Mentor 6 developers through code reviews, pair programming, and knowledge sharing; run internal workshops and maintain up-to-date documentation.
- Collaborate with product and platform teams to scope use cases, set metrics, control costs, and ship reliable model updates on a regular cadence.
- Drove Scrum delivery (grooming, sprint reviews) while documenting versioned changes for stakeholders.

Lead Product Engineer (Contract) | Edubanc (Sterling Bank Subsidiary) | Lagos, Nigeria | May 2024 – Present

- Architected event-driven payout and wallet flows in Node.js/TypeScript with idempotency, retries, and DLQs; stabilized partner timeouts with exponential backoff and jitter.

- Built low-latency APIs for loan decisions, disbursements, and repayments; enforced rate limits and request budgets to protect core services.
- Integrated payment providers with signed webhooks and replay protection; added audit logs and runbooks for production recovery.
- Deployed on AWS (API Gateway, Lambda, SQS, CloudWatch); added dashboards for queue lag, p95 latency, and error budgets; led on-call and incident reviews.
- Introduced WebSocket and webhook patterns for near-real-time status events to ops dashboards and clients.
- Worked across SQL and NoSQL data stores (including MongoDB) for transaction data, configuration, and audit trails; improved transaction efficiency by ~40% through indexing, caching, and removing slow paths.
- Drove platform modernization with containerized services (Docker/Kubernetes), CI/CD, change control, and DevSecOps checks (static analysis, dependency scanning, secrets hygiene).
- Set and monitored performance targets for remittance endpoints (p95 latency, throughput, error budgets); led incident response and RCA with corrective actions.
- Produced and maintained architecture diagrams, API catalogs, and operational runbooks; mentored engineers and coordinated with Product, Compliance, and Operations on corridor launches and partner certifications.
- Supported mobile-facing integrations by designing backend contracts consumed by mobile apps, ensuring stable versioning and predictable state transitions.

Software Engineer (Full Stack) | Heirs Holdings | Lagos, Nigeria | Oct 2023 – Nov 2024

- Optimized Converge ERP workflows for 40,000 employees across a \$6.9B portfolio, improving processing efficiency by 50% through backend performance tuning, SQL query optimization, caching, and workflow refactors.
- Designed and delivered ERP workflow modules as Java/Spring-based services with well-defined REST APIs; produced conceptual and detailed technical designs aligned to enterprise standards.
- Built event-driven processing for workflow and notification pipelines using RabbitMQ to improve reliability, ordering, and async job throughput for high-volume internal operations.
- Worked with Service Management and Operations to plan releases, monitor system health, and manage incidents and post-release fixes; introduced runbooks and measurable operational checks.
- Developed the TEF Connect platform supporting 50,000+ entrepreneur registrations and pitch sessions, using scalable backend services and secure API integrations.
- Streamlined 13th-month allowance processing by implementing strict validation rules, consistent state handling, and automated reconciliation, increasing accuracy to 99.99% and reducing manual effort by 60%.
- Implemented a gamification module backed by rule-based scoring services and persistent storage (SQL/NoSQL where appropriate), increasing idea generation by 30%.
- Automated leave allowance processing with backend workflows and scheduled jobs, reducing admin workload by 75% and improving turnaround time.
- Delivered Node.js services with Redis caching and queue-backed jobs; improved payroll/ERP flows under peak load with batching and idempotent writes.
- Led incident response: added alerts, error budgets, and RCAs; reduced repeat outages via guardrails and playbooks.

Solution Engineer (Full Stack) | DipoleDIAMOND | Lagos, Nigeria | Jan 2023 – Nov 2023

- Designed and shipped bank-integrated backend services and chat/payment workflows, producing solution designs aligned to enterprise architecture and security standards; implemented core services in Java/Spring with well-structured REST APIs.
- Led a team of 5 to build chatbot solutions for tier-one and tier-two banks and 4 corporate clients, delivering secure integrations and reliable production rollouts.
- Integrated vendor and bank APIs, enforcing authentication, authorization, and audit logging; worked closely with Security and compliance teams to meet regulatory requirements.
- Built asynchronous processing for payment and notification workflows using Kafka where appropriate, improving resiliency and reducing failures during peak usage.
- Worked across SQL and NoSQL databases (MongoDB) for conversation state, event logs, and operational reporting, optimizing performance with indexing and caching.
- Coordinated test plans and supervised UAT; supported go-live transitions with runbooks, training notes, and

performance tuning.

- Managed a team of 3 to deliver an AI-powered video conferencing platform with transcription features; designed backend services and integrations to support real-time workloads and secure access.
- Standardized deployments using containerization (Docker/Kubernetes where applicable) and CI/CD practices to improve release consistency and reduce rollback risk.

Software Engineer | The People Practice (Contract) | Remote (Lagos, Nigeria) | Oct 2023 – Dec 2023

- Designed and implemented 8 APIs to optimize HR Tech solution, improving search functionality and system performance by 35%.

Software Engineer (Full Stack) | ZEDIRAX MULTI CONCEPT | Lagos, Nigeria | Jan 2019 – Dec 2022

- Directed a 15-member development team to deliver a health-tech product on time and within budget.
- Spearheaded end-to-end implementation of SaaS offerings, achieving seamless client integrations and adoption.
- Refactored and developed reusable UI components, improving design consistency and boosting user satisfaction by 70%.
- Identified and remediated a critical security vulnerability, reducing incidents by 50%.
- Architected and maintained websites for Edo State Specialist Hospital and Acaderax, and provided technical support for Ciuci Consulting, enhancing site performance and user engagement.

EDUCATION

Bachelor of Science, Biochemistry | University of Benin, Edo, Nigeria | 2013 – 2017

CERTIFICATIONS

Introducing Generative AI with AWS (UDACITY) | 2025

AWS Educate Machine Learning Foundations (AWS) | 2025

Amazon Elastic Container Service (AWS ECS) | 2023

Google Africa Developer Training Program | 2019

PERSONAL PROJECTS

- [DipoleVibe](#) (2023): Video-sharing and AI meeting recording app. Tech: NodeJS, ReactJS, .NET, Python, WebRTC, PostgreSQL, MongoDB, Microservices.
- [Converge ERP](#) (2024): ERP solution managing business processes for Heirs Holdings. Tech: ReactJS, .NET, NodeJS, MySQL, Azure VM, Tailwind, ChartJS.
- [Edubanc](#) (2025): Web application offering flexible financing solutions. Tech: NextJS, NestJS, PostgreSQL, Python.